WORKED SOLUTION

STEP III 2016, Question 10

Mechanics • Circular Motion on a Plane

STEP III 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q10

The key idea

While the string is taut and the particle stays on the plane, P is always a distance a from A and lies in the plane, so it moves on a circle *in the plane*, centred at the foot of the perpendicular from A. Within the plane this is just a vertical circle under a reduced gravity $g\sin\alpha$; out of the plane, the string pulls P away from the surface, and the normal reaction must stay positive. So we find the tension, find the reaction, and ask where each is smallest.

Solution

Setting up

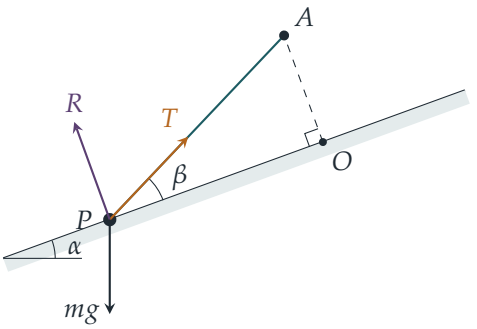
Model P as a particle, the string as light and inextensible, and the plane as smooth. Let O be the foot of the perpendicular from A to the plane. Since the string has length a and makes angle β with the plane,

$$AO = a\sin\beta, \qquad OP = a\cos\beta.$$

In equilibrium the only force with a component along the plane besides gravity is the tension, so the string must lie along a line of greatest slope, with P directly down the slope from O. Resolving along and perpendicular to the plane (tension T_0, reaction R_0):

$$T_0\cos\beta = mg\sin\alpha, \qquad R_0 + T_0\sin\beta = mg\cos\alpha,$$

so $R_0 = mg(\cos\alpha - \sin\alpha\tan\beta) = mg\cos(\alpha+\beta)/\cos\beta$. For the equilibrium to exist at all we therefore need $\cos(\alpha+\beta) > 0$.



The figure is the cross-section through a line of greatest slope at the start. AO is perpendicular to the plane.

Once the impulse is applied, as long as the string stays taut and P stays on the plane, $AP = a$ and $AO = a\sin\beta$, so by Pythagoras $OP = a\cos\beta$ always: P moves on a circle of radius

$$r = a\cos\beta$$

in the plane, centred at O. The string always makes angle β with the plane, and its projection onto the plane points from P to O. The impulse is horizontal and parallel to the plane, so it is tangent to this circle at its lowest point.

Let θ be the angle OP makes with the downward line of greatest slope through O, so $\theta = 0$ at the start, and let v be the speed. The weight has component $mg\sin\alpha$ down the slope (in the plane) and $mg\cos\alpha$ into the plane.

Perpendicular to the plane P has no acceleration, so

$$R + T\sin\beta = mg\cos\alpha. \tag{1}$$

Towards O, within the plane, the acceleration is v^2/r. The tension contributes $T\cos\beta$ and the down-slope weight component contributes $-mg\sin\alpha\cos\theta$:

$$T\cos\beta - mg\sin\alpha\cos\theta = \frac{mv^2}{a\cos\beta}. \tag{2}$$

Energy: the reaction and tension do no work (each is perpendicular to the velocity), and P rises a distance $r(1-\cos\theta)$ along the slope, i.e. a height $r(1-\cos\theta)\sin\alpha$. Hence

$$v^2 = u^2 - 2ga\cos\beta\sin\alpha\,(1-\cos\theta). \tag{3}$$

Not leaving the plane immediately

Just after the impulse, $\theta = 0$ and $v = u$. From (2), $T\cos\beta = mg\sin\alpha + \dfrac{mu^2}{a\cos\beta}$, and then (1) gives

$$\begin{aligned} R &= mg\cos\alpha - \tan\beta\left(mg\sin\alpha + \frac{mu^2}{a\cos\beta}\right) \\ &= \frac{m}{\cos\beta}\left(g\cos\alpha\cos\beta - g\sin\alpha\sin\beta\right) - \frac{mu^2\tan\beta}{a\cos\beta} \\ &= \frac{m}{a\cos\beta}\Big(ag\cos(\alpha+\beta) - u^2\tan\beta\Big). \end{aligned}$$

Since $\cos\beta > 0$, the reaction is positive, and the particle does not immediately leave the plane, if

$$ag\cos(\alpha+\beta) > u^2\tan\beta,$$

as required.

A complete circle

Now use (3) in (2) to get the tension at a general position:

$$T\cos\beta = \frac{mu^2}{a\cos\beta} - 2mg\sin\alpha + 3mg\sin\alpha\cos\theta,$$

and so, from (1),

$$R = mg\cos\alpha - \tan\beta\left(\frac{mu^2}{a\cos\beta} - 2mg\sin\alpha + 3mg\sin\alpha\cos\theta\right).$$

Ask yourself Where on the circle is the particle most likely to leave the plane: at the top or at the bottom?

Intuition from a vertical circle says "the top", but here that's the danger point for the *string*, not the plane. The string pulls P away from the plane with force $T\sin\beta$, and the tension is greatest at the bottom, where P is moving fastest and gravity in the plane also pulls away from O. Indeed R is smallest when $\cos\theta$ is largest, i.e. at $\theta = 0$, and T is smallest when $\cos\theta = -1$, at the top. So for a complete circle in contact we need two things.

- **Contact at the bottom.** $R > 0$ at $\theta = 0$, which is exactly the condition of the first part: $u^2\tan\beta < ag\cos(\alpha+\beta)$.
- **String taut at the top.** $T \geq 0$ at $\theta = \pi$: $\dfrac{u^2}{a\cos\beta} - 5g\sin\alpha \geq 0$, i.e. $u^2 \geq 5ag\sin\alpha\cos\beta$. (This also makes $v^2 > 0$ at the top, by (3).)

Both can hold only if, multiplying the second by $\tan\beta > 0$,

$$5ag\sin\alpha\cos\beta\tan\beta \le u^2\tan\beta < ag\cos(\alpha+\beta),$$

that is,

$$5\sin\alpha\sin\beta < \cos\alpha\cos\beta - \sin\alpha\sin\beta \iff 6\sin\alpha\sin\beta < \cos\alpha\cos\beta.$$

Dividing by $\cos\alpha\cos\beta > 0$ gives the necessary condition

$$6\tan\alpha\tan\beta < 1.$$

(It is also sufficient: if it holds, any u with $5ag\sin\alpha\cos\beta \le u^2 < ag\cos(\alpha+\beta)/\tan\beta$ works, because R and T are then positive everywhere on the circle.)

Checks

- With $u = 0$ the reaction from the first part becomes $mg\cos(\alpha+\beta)/\cos\beta$, matching the equilibrium value R_0.
- If $\beta \to 0$ (string almost along the plane) the string cannot lift the particle, and the condition $6\tan\alpha\tan\beta < 1$ holds automatically, as it should.
- With $\alpha = 30°$, $\beta = 5°$ (so $6\tan\alpha\tan\beta \approx 0.30$), $a = 1\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, the allowed range is $24.4 \le u^2 < 91.8$ (in $\text{m}^2\,\text{s}^{-2}$). Taking $u^2 = 50$ gives a smallest reaction of about $0.37mg$ at the bottom and a smallest tension of about $2.63mg$ at the top. We confirmed by computer, by solving Newton's second law in three dimensions all round the circle, that R and T stay positive, with the values above.

Common slips

- Using the full string length a as the radius of the circle: the radius is $a\cos\beta$.
- Assuming the particle is most likely to leave the plane at the top of the circle. The reaction is smallest at the bottom.
- Using g instead of $g\sin\alpha$ in the energy equation: the particle rises only $r(1-\cos\theta)\sin\alpha$ vertically.

Beyond STEP

This is motion under constraints: the plane and the string each supply a reaction force whose size is whatever is needed to keep the constraint, and the motion stays valid only while those forces have the right sign. In a first-year course you'd learn to handle this systematically, using Lagrangian mechanics to remove the constraints and recover the reactions afterwards. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems of exactly this "when does it lose contact?" kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.